Q.1 $${\text{In }}{\mathbb{R}^2},{\text{Linearly dependent vectors lie on the same}} - - - - - .$$

A. Plane

B. Line

Q.2 $Let{\kern 1pt} {\kern 1pt} {\kern 1pt} {\kern 1pt} {\kern 1pt} T:{R^2} \to {R^2}{\kern 1pt} {\kern 1pt} {\kern 1pt} be{\kern 1pt} {\kern 1pt} {\kern 1pt} {\kern 1pt} transformation{\kern 1pt} {\kern 1pt} {\kern 1pt} T({x_1},{x_2}) = (x,0).{\kern 1pt} {\kern 1pt} {\kern 1pt} The{\kern 1pt} {\kern 1pt} {\kern 1pt} null{\kern 1pt} {\kern 1pt} {\kern 1pt} space{\kern 1pt} {\kern 1pt} (or{\kern 1pt} {\kern 1pt} \ker nel)\,N(T)\,\,of\,T\,\,is$

A. $(0,1)$

B. $(x_1,0)$

C. $(0, x_2)$

D. $(1, 0)$

Q.3 We can add the matrices of _____.

A. same number of columns

B. same number of rows.

C. same order

D. different order

Q.4 $${\text{Which of the following system will have the trivial solution?}}$$

A. $$x = y = 0$$

B. $$x = 0, y = 1$$

C. $$x = y = 1$$

D. $$x = 1, y = 0$$

Q.5 $${(AB)^t} = ?$$

A. $${B^t}{A^t}$$

B. BA

Q.6 Two vectors are linearly dependent if and only if they lie_____

A. on a line parallel to x-axis

B. on the same line through origin

Q.7 $$\begin{gathered} {\text{What will be the order of matrix:}} A - B{\text{, if the orders of both}} ~ \ A ~{\text{and}} ~B ~{\text{is}}~ 2 \times 3? \\ \end{gathered}$$

A. $$2 3$$

B. $$0 0$$

C. $$2 2$$

D. $$3 3$$

Q.8 A system of linear equations is said to be homogeneous if it can be written in the form _____.

A. AX = 0

B. AX = B

Q.9 $$\begin{array}{*{20}{c}} {\text{The system of equations}:} \\ {{x_1} = 1, } \\ \begin{gathered} {x_2} = 0; \ {\text{can be expressed in the form - - - - - }}. \\ \end{gathered} \end{array}$$

A. $${array}{*{20}{c}} {Ax = 0} \\ {By = 1} {array}$$

B. $${array}{*{20}{c}} {Ax = 1} \\ {By = 0} {array}$$

C. $$Ax = 0$$

D. $$Ax = b$$

Q.10 Linear algebra is the study of linear sets of equations and their ---------- properties.

A. transformation

B. non-linear

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